

STUDENT ACTIVITY

Comparative Evaluation — Activity

Activity: Evaluating Classification Model Performance

Q: Read the scenario and the two model evaluation reports below. Evaluate both models by analyzing their performance across the four classification evaluation metrics: Accuracy, Precision, Recall, and F1 Score.

Next, based on the provided text, justify which model is the superior choice for this specific enterprise system in four sentences. Ensure you explicitly explain why relying solely on Accuracy is dangerous for this dataset and how the F1 Score provides a more balanced evaluation.

The Scenario: A banking institution is deploying a supervised Machine Learning classification model for Fraud Detection. The dataset is highly imbalanced because fraudulent transactions represent a very small percentage of the total data compared to legitimate transactions. The team has trained two different algorithms and must choose one for production.

- **Model A (Logistic Regression):** The evaluation report shows an Accuracy of 99%, meaning a high percentage of overall correct predictions. However, the Confusion Matrix reveals a high number of False Negatives. The model has a low Recall (it fails to correctly identify many actual positive fraud cases) and a low F1 Score.
- **Model B (Random Forest):** The evaluation report shows a slightly lower Accuracy of 95%. However, because it combines multiple Decision Trees to reduce overfitting, it maintains a high Precision and high Recall. The Confusion Matrix shows very few False Negatives, resulting in a high F1 Score on the imbalanced dataset.

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Evaluation of the Two Models

Evaluation Metric	Model A – Logistic Regression	Model B – Random Forest
Accuracy	99% – Very high overall accuracy, but can be misleading because the dataset is highly imbalanced.	95% – Slightly lower accuracy, but still provides strong overall performance.
Precision	Lower precision because the model makes less reliable fraud predictions.	High Precision – Most transactions predicted as fraud are actually fraudulent, reducing false alarms.
Recall	Low Recall – Misses many actual fraud cases due to a high number of False Negatives .	High Recall – Correctly identifies most fraud cases with very few False Negatives .
F1 Score	Low F1 Score because both Precision and Recall are poor.	High F1 Score because it maintains both high Precision and high Recall, making it well suited for imbalanced datasets.

Comparison of the Models

Model A – Logistic Regression

- **Accuracy:** Very high (99%).
- **Weakness:** Low Recall causes many fraudulent transactions to go undetected.
- **Confusion Matrix:** High number of **False Negatives**.
- **Overall:** Not suitable for fraud detection because missing fraud cases can lead to significant financial losses.

Model B – Random Forest

- **Accuracy:** Slightly lower (95%).
- **Strengths:** High Precision and High Recall.
- **Confusion Matrix:** Very few **False Negatives**, meaning most fraud cases are detected.
- **Overall:** More reliable and robust for fraud detection on an imbalanced dataset.

Best Model for the Enterprise System

Model B (Random Forest) is the superior choice because it achieves **high Precision, high Recall, and a high F1 Score**, which are more important than accuracy alone for fraud detection. Relying only on **Accuracy** is dangerous because an imbalanced dataset can produce a high accuracy even when the model fails to detect many actual fraud cases. The **high F1 Score** of Model B provides a balanced evaluation by considering both Precision and Recall, ensuring that fraud cases are detected while minimizing incorrect predictions. Therefore, **Random Forest** is

the better production model because it reduces False Negatives, improves fraud detection, and performs more reliably on imbalanced data.

Performance Evaluation

Metric	Model A (Logistic Regression)	Model B (Random Forest)
Accuracy	99% (Very High)	95% (High)
Precision	Low	High
Recall	Low (Many False Negatives)	High (Very Few False Negatives)
F1 Score	Low	High

Conclusion

Model B (Random Forest) is the better model for the banking fraud detection system. Although **Model A** has a higher **Accuracy (99%)**, accuracy alone is misleading on highly **imbalanced datasets** because it may hide poor fraud detection performance. **Model B** achieves a **high F1 Score** by balancing **Precision** and **Recall**, making it much more effective at identifying fraudulent transactions. Therefore, **Random Forest** is the preferred choice because it detects more fraud cases, minimizes False Negatives, and provides a more reliable evaluation for real-world banking applications.